

Case Study #2: Forecasting Walmart's Revenue

Case Solutions

The data set for case study #2 represents quarterly revenues (in \$million) in Walmart from the first quarter of 2006 through the third quarter (Q3) of 2025 (*673_case2.csv*). This quarterly data is collected from www.macrotrends.net/stocks/charts/WMT/walmart/revenue. The goal is to forecast Walmart's quarterly revenue in Q4 of 2025 and four quarters (Q1-Q4) of 2026-2027.

Questions

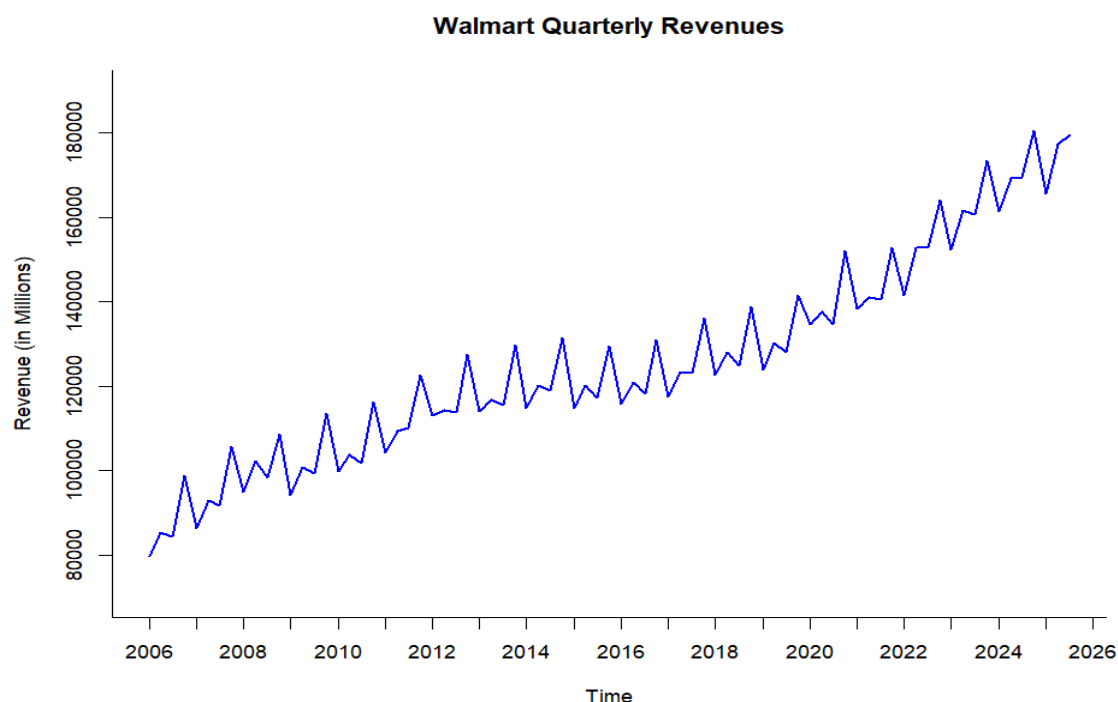
1. Plot the data and visualize time series components.

1a. Create time series data set in R using the `ts()` function.

(The answers to this question can vary). We loaded into R, using the `read.csv()` function, the *673_case2.csv* data, and then created a time series data *revenue.ts* for Walmart's revenue using the `ts()` function from Q1 of 2006 through Q3 of 2025, with the frequency of 4 for quarterly data.

1b. Apply the `plot()` function to create a data plot with the historical data, provide it in your report, and explain what time series components can be visualized in this plot.

The plot of Walmart's revenue is shown below.



The plot shows potentially an upward non-linear (but also possibly linear) trend pattern. The plot also

shows an additive seasonal pattern with the high revenue at the end of each year (Q4) and low revenue at the beginning of the year (approximately in Q1-Q2).

2. Apply three regression models using data partition.

Consider the following 3 regression-based models:

- i. Regression model with seasonality
- ii. Regression model with linear trend and seasonality
- iii. Regression model with quadratic trend and seasonality.

2a. Develop data partition with the validation partition of 19 periods and the rest for the training partition.

The training period (*train.ts*) and validation period (*valid.ts*) time series partitions of 60 and 19 records, respectively, are listed below (this is optional).

train.ts

	Qtr1	Qtr2	Qtr3	Qtr4
2006	79676	85430	84467	98795
2007	86410	92999	91865	105749
2008	94940	102342	98345	108627
2009	94242	100876	99373	113594
2010	99811	103726	101952	116360
2011	104189	109366	110226	122728
2012	113010	114282	113800	127559
2013	114070	116830	115688	129706
2014	114960	120125	119001	131565
2015	114826	120229	117408	129667
2016	115904	120854	118179	130936
2017	117542	123355	123179	136267
2018	122690	128028	124894	138793
2019	123925	130377	127991	141671
2020	134622	137742	134708	152079

valid.ts

	Qtr1	Qtr2	Qtr3	Qtr4
2021	138310	141048	140525	152871
2022	141569	152859	152813	164048
2023	152301	161632	160804	173388
2024	161508	169335	169588	180554
2025	165609	177402	179496	

2b. Use the *tslm()* function for the training partition to develop each of the 3 regression models from the above list. Apply the *summary()* function to identify the model structure and parameters for each regression model, show them in your report, and also present the respective model equation and define its predictors. Briefly explain if the model is a good fit, statistically significant, and thus may be applied for forecasting. Use each model to forecast revenues for the validation period using the *forecast()* function, and present this forecast in your report

We developed the 3 regression models identified in question 2.

1. Regression Model with Seasonality

The model using the *summary()* function is presented below:

```
call:
tslm(formula = train.ts ~ season)

Residuals:
    Min       1Q   Median       3Q      Max
-29045 -11575   4080   9012  26473
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	108721	3768	28.854	< 2e-16 ***
season2	5050	5329	0.948	0.34740
season3	3351	5329	0.629	0.53205
season4	16885	5329	3.169	0.00248 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14590 on 56 degrees of freedom
Multiple R-squared: 0.1693, Adjusted R-squared: 0.1248
F-statistic: 3.803 on 3 and 56 DF, p-value: 0.01493

The *regression model with seasonality* contains three predictors: seasonal dummy variables for Q2 (season2 – D_2), Q3 (season3 – D_3) and Q4 (season4 – D_4). The model's equation is:

$$y_t = 108721 + 5050 D_2 + 3351 D_3 + 16885 D_4$$

The model's summary shows a low (weak) R-squared of 0.1693 and even lower adjusted R_squared of 0.1248, and two regression coefficients for the seasonal variables are statistically insignificant. However, the coefficient for D_4 will be statistically significant for p-value < 0.01. Overall, this regression model is not a good fit, and thus cannot be applied for time series forecasting.

2. Regression Model with Linear Trend and Seasonality

The model using the *summary()* function is presented below:

Call:

```
tslm(formula = train.ts ~ trend + season)
```

Residuals:

Min	1Q	Median	3Q	Max
-6812.5	-2446.3	-381.9	3060.1	7465.0

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	85694.43	1138.94	75.241	< 2e-16 ***
trend	794.02	25.24	31.455	< 2e-16 ***
season2	4255.58	1234.17	3.448	0.00109 **
season3	1762.55	1234.95	1.427	0.15917
season4	14503.19	1236.23	11.732	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3379 on 55 degrees of freedom
Multiple R-squared: 0.9563, Adjusted R-squared: 0.9531
F-statistic: 300.6 on 4 and 55 DF, p-value: < 2.2e-16

The regression model contains 4 predictors: trend index (t) and 3 seasonal dummy variables for Q2 (season2 – D_2), Q3 (season3 – D_3) and Q4 (season4 – D_4). The model's equation is:

$$y_t = 85694.43 + 794.02 t + 4255.58 D_2 + 1762.55 D_3 + 14503.19 D_4$$

The model's summary shows a very high R-squared of 0.9563 and adj. R-squared of 0.9531, which is a very good fit for the training data, and the regression coefficients for t , D_2 , and D_4 are statistically significant. Although the regression coefficient for D_3 is statistically insignificant for p-value > 0.05, overall, this regression model is a very good fit and statistically significant, and thus can be applied for time series forecasting.

The forecast for the training period is the following:

	Point	Forecast	Lo 0	Hi 0
2021 Q1		134129.9	134129.9	134129.9
2021 Q2		139179.5	139179.5	139179.5
2021 Q3		137480.5	137480.5	137480.5
2021 Q4		151015.2	151015.2	151015.2
2022 Q1		137306.0	137306.0	137306.0
2022 Q2		142355.6	142355.6	142355.6
2022 Q3		140656.6	140656.6	140656.6
2022 Q4		154191.3	154191.3	154191.3
2023 Q1		140482.1	140482.1	140482.1
2023 Q2		145531.7	145531.7	145531.7
2023 Q3		143832.7	143832.7	143832.7
2023 Q4		157367.4	157367.4	157367.4
2024 Q1		143658.2	143658.2	143658.2
2024 Q2		148707.8	148707.8	148707.8
2024 Q3		147008.8	147008.8	147008.8
2024 Q4		160543.5	160543.5	160543.5
2025 Q1		146834.3	146834.3	146834.3
2025 Q2		151883.9	151883.9	151883.9
2025 Q3		150184.9	150184.9	150184.9

3. Regression Model with Quadratic Trend and Seasonality

The model using the *summary()* function is presented below:

Call:

```
tslm(formula = train.ts ~ trend + I(trend^2) + season)
```

Residuals:

Min	1Q	Median	3Q	Max
-4914.3	-1858.3	-600.4	1937.2	7673.1

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	81899.443	1338.764	61.175	< 2e-16 ***
trend	1161.865	89.516	12.979	< 2e-16 ***
I(trend^2)	-6.030	1.422	-4.240	8.8e-05 ***
season2	4243.516	1078.843	3.933	0.000241 ***
season3	1750.491	1079.520	1.622	0.110726
season4	14503.194	1080.643	13.421	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2954 on 54 degrees of freedom

Multiple R-squared: 0.9672, Adjusted R-squared: 0.9641

F-statistic: 318.3 on 5 and 54 DF, p-value: < 2.2e-16

The *regression model with quadratic trend and seasonality* contains 5 predictors: trend index (t), squared trend index (t^2), and 3 seasonal dummy variables for Q2 (season2 – D_2), Q3 (season3 – D_3) and Q4 (season4 – D_4). The model's equation is:

$$y_t = 81899.44 + 1161.87 t - 6.03 t^2 + 4243.52 D_2 + 1750.49 D_3 + 14503.19 D_4$$

The model summary shows a very high R-squared of 0.9672 and adj. R-squared of 0.9641, and regression coefficients for t , t^2 , D_2 , and D_4 are statistically significant. Although the regression coefficient for D_3 is statistically insignificant for p-value > 0.05, overall, this regression model is a very good fit and statistically significant, and thus can be applied for time series forecasting..

The forecast for the training period is the following:

	Point	Forecast	Lo 0	Hi 0
2021 Q1		130334.9	130334.9	130334.9
2021 Q2		134998.6	134998.6	134998.6
2021 Q3		132913.7	132913.7	132913.7
2021 Q4		146062.4	146062.4	146062.4
2022 Q1		131943.2	131943.2	131943.2
2022 Q2		136558.6	136558.6	136558.6

2022 Q3	134425.4	134425.4	134425.4
2022 Q4	147525.9	147525.9	147525.9
2023 Q1	133358.4	133358.4	133358.4
2023 Q2	137925.6	137925.6	137925.6
2023 Q3	135744.2	135744.2	135744.2
2023 Q4	148796.5	148796.5	148796.5
2024 Q1	134580.8	134580.8	134580.8
2024 Q2	139099.7	139099.7	139099.7
2024 Q3	136870.1	136870.1	136870.1
2024 Q4	149874.1	149874.1	149874.1
2025 Q1	135610.1	135610.1	135610.1
2025 Q2	140080.8	140080.8	140080.8
2025 Q3	137802.9	137802.9	137802.9

2c. Apply the *accuracy()* function to compare performance measures of the 3 regression models you developed in 2b. Present the accuracy measures in your report, compare them, and, using MAPE and RMSE, identify the *two most accurate regression models for forecasting*.

The accuracy measures for the 3 regression models are presented below:

```
> Regression Model with Seasonality
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 45285.07 46902.59 45285.07 27.968 27.968 0.828    4.926

> Regression Model with Linear Trend and Seasonality
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 13858.43 16006.32 13858.43  8.356  8.356 0.795    1.66

> Regression Model with Quadratic Trend and Seasonality
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 21639.69 24071.78 21639.69 13.135 13.135 0.814    2.501
```

Based on the lowest values of MAPE and RMSE accuracy measures for the validation period, the best model is *regression model with linear trend and seasonality* (MAPE and RMSE for the validation period forecast are 8.36% and 16006.32, respectively). This model was selected as the first most accurate model for forecasting. The second model that was selected is the *regression model with quadratic trend and seasonality*. It has the second lowest MAPE of 13.14% and RMSE of 24071.78.

3. Employ the entire data set to make time series forecast.

3a. Apply the two most accurate regression models identified in 2c to make the forecast in Q4 of 2025 and the four quarters (Q1-Q4) of 2026-2027. For that, use the entire data set to develop the regression model using the *tslm()* function. Apply the *summary()* function to identify the model structure and parameters, show them in your report, and also present the respective model equation and define its predictors. Briefly explain if the model is a good fit, statistically significant, and thus may be applied for forecasting. Use each model to forecast Walmart's revenue in Q4 of 2025 and Q1-Q4 of 2026-2027 using the *forecast()* function, and present this forecast in your report.

1. Regression Model with Linear Trend and Seasonality

For forecasting Walmart's revenue in Q4 of 2025 and Q1-Q4 of 2026-2027, we apply for the entire data set the *regression model with linear trend and seasonality*, which was identified to be the first most accurate model in question 2c. The *regression model with linear trend and seasonality* for the entire data set is presented below:

```
call:
tslm(formula = revenue.ts ~ trend + season)
```

```

Residuals:
    Min       1Q   Median       3Q      Max
-10299  -5671   1115   3363  17077

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 80296.70   1798.54  44.645 < 2e-16 ***
trend       1005.36     30.09  33.413 < 2e-16 ***
season2      4930.79   1927.79   2.558  0.0126 *
season3      2698.68   1928.50   1.399  0.1659
season4     13960.36   1952.99   7.148  5.2e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6095 on 74 degrees of freedom
Multiple R-squared:  0.9409, Adjusted R-squared:  0.9377
F-statistic: 294.4 on 4 and 74 DF,  p-value: < 2.2e-16

```

This *regression model with linear trend and seasonality* contains 4 predictors: trend index (t) and 3 seasonal dummy variables for Q2 (season2 – D₂), Q3 (season3 – D₃) and Q4 (season4 – D₄). The regression equation is:

$$y_t = 80296.70 + 1005.36 t + 4930.79 D_2 + 2698.68 D_3 + 13960.36 D_4$$

The model's summary shows a high R-squared of 0.9409 and adj. R-squared of 0.9377, and all regression coefficients are statistically significant except for the regression coefficient for D₃. Overall, this regression model is a very good fit and can be used for forecasting Walmart's revenue in Q4 of 2025 and Q1-Q4 of 2026-2027. The respective forecast is shown below:

	Point	Forecast	Lo 0	Hi 0
2025 Q4		174685.8	174685.8	174685.8
2026 Q1		161730.8	161730.8	161730.8
2026 Q2		167666.9	167666.9	167666.9
2026 Q3		166440.2	166440.2	166440.2
2026 Q4		178707.2	178707.2	178707.2
2027 Q1		165752.2	165752.2	165752.2
2027 Q2		171688.4	171688.4	171688.4
2027 Q3		170461.6	170461.6	170461.6
2027 Q4		182728.7	182728.7	182728.7

2. Regression Model with Quadratic Trend and Seasonality

For forecasting Walmart's revenue in Q4 of 2025 and Q1-Q4 of 2026-2027, we apply for the entire data set the *regression model with quadratic trend and seasonality*, which was identified to be the second best model in question 2c. The *regression model with quadratic trend and seasonality* for the entire data set is presented below:

```

Call:
tslm(formula = revenue.ts ~ trend + I(trend^2) + season)

Residuals:
    Min       1Q   Median       3Q      Max
-9115.8 -4109.7   299.7  3893.7  9914.2

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 88385.286   1931.622  45.757 < 2e-16 ***
trend       398.904     98.478   4.051 0.000126 ***
I(trend^2)    7.581      1.193   6.355 1.60e-08 ***
season2      4938.372   1557.392   3.171 0.002222 **
season3      2698.682   1557.961   1.732 0.087463 .
season4     14362.138   1579.012   9.096 1.23e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4924 on 73 degrees of freedom
Multiple R-squared:  0.9619, Adjusted R-squared:  0.9593
F-statistic: 369 on 5 and 73 DF,  p-value: < 2.2e-16

```

This *regression model with quadratic trend and seasonality* contains 5 predictors: trend index (t), squared trend index (t^2), and 3 seasonal dummy variables for Q2 (season2 – D_2), Q3 (season3 – D_3) and Q4 (season4 – D_4). The regression equation is:

$$y_t = 88385.29 + 398.90 t + 7.58 t^2 + 4938.37 D_2 + 2698.68 D_3 + 14362.14 D_4$$

The model's summary shows a very high R-squared of 0.9557 and adj. R-squared of 0.9528, and all regression coefficients are statistically significant with exception for the regression coefficient for D_3 . Overall, this regression model is a very good fit for the historical data set and statistically insignificant, and thus may be a good model for forecasting Walmart's revenue in Q4 of 2025 and Q1-Q4 of 2026-2027. The appropriate forecast is shown below:

	Point Forecast	Lo 0	Hi 0
2025 Q4	183176.1	183176.1	183176.1
2026 Q1	170433.4	170433.4	170433.4
2026 Q2	177006.3	177006.3	177006.3
2026 Q3	176416.4	176416.4	176416.4
2026 Q4	189744.7	189744.7	189744.7
2027 Q1	177062.6	177062.6	177062.6
2027 Q2	183696.2	183696.2	183696.2
2027 Q3	183166.8	183166.8	183166.8
2027 Q4	196555.8	196555.8	196555.8

3b. Apply the *accuracy()* function to compare the performance measures of the regression models developed in 3a with those for the naïve and seasonal naïve forecasts. Present the accuracy measures in your report, compare them, and identify, using MAPE and RMSE, which forecast is most accurate to forecast Walmart's quarterly revenue in Q4 of 2025 and Q1-Q4 of 2026-2027.

The accuracy measures for the three forecasts specified in question 3b are listed below along with the accuracy for the respective naïve and seasonal naïve forecast:

```
> Regression Model with Linear Trend and Seasonality
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set  0 5899.425 4850.481 -0.115  3.699  0.887      0.541

> Regression Model with Quadratic Trend and Seasonality
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set  0 4733.615 4033.899 -0.15  3.328  0.863      0.484

> Naïve Model
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 1279.744 9796.452 8260.795 0.721  6.677 -0.718      1

> Seasonal Naïve Model
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 4729.24 5882.997 4883.613 3.652  3.789  0.735      0.555
```

Based on the lowest MAPE of 3.33%, the best model is the *regression model with quadratic trend and seasonality*. For RMSE, the *regression model with quadratic trend and seasonality* has also the lowest value of 4733.62. Therefore, the recommendation is to apply the *regression model with quadratic trend and seasonality* for forecasting Walmart's revenue in Q4 of 2025 and Q1-Q4 of 2026-2027.